

Project Carrier Course, Part 1 CM2015

Lecture 1

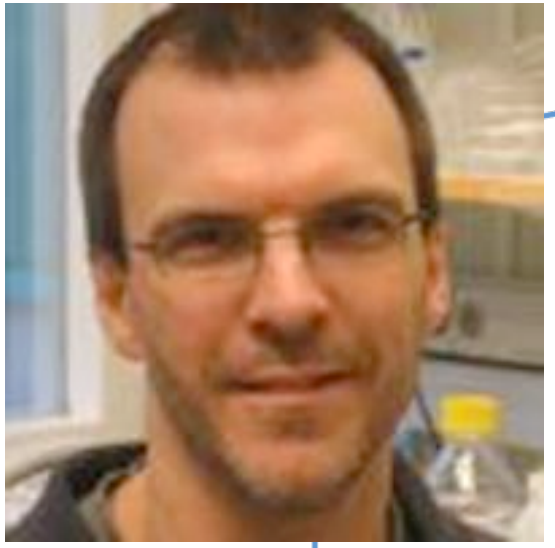
Autumn 2025 - Martin Jacobsson

People

- Martin Jacobsson, marjacob@kth.se, room 7507
- Massimiliano Colarieti-Tosti, mct@kth.se
- Philip Köck, koeck@kth.se







People, more

- Linus Remahl, MakerSpace Supervisor

- TAs:
  - Nada Chaari, PostDoc
  - Shucheng Yi, MSc student, Medical Engineering, Year 2
  - Max Gulda, MSc student, Electrical engineering, Year 2
  - Alberto Crivelli, MSc student, Medical Engineering, Year 2+

Course content:

Part 1: Lab series on:

- Developing a heart rate sensor connected to EHR (Electronic Health Record) system
- Electronics, programming, DSP, EHR systems, User interfaces, Statistics

Part 2: Project

- Project Planning
- Group Dynamics

Next year (PCC2 and PCC3): More projects

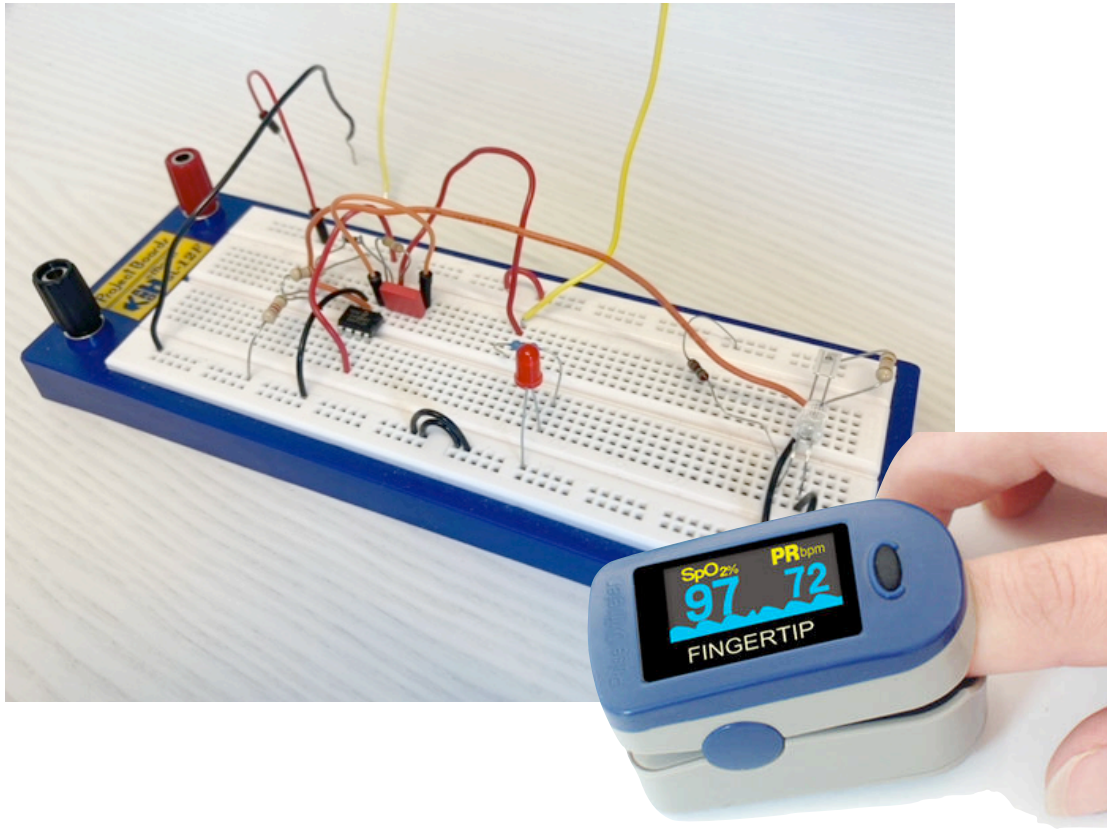
Period 2

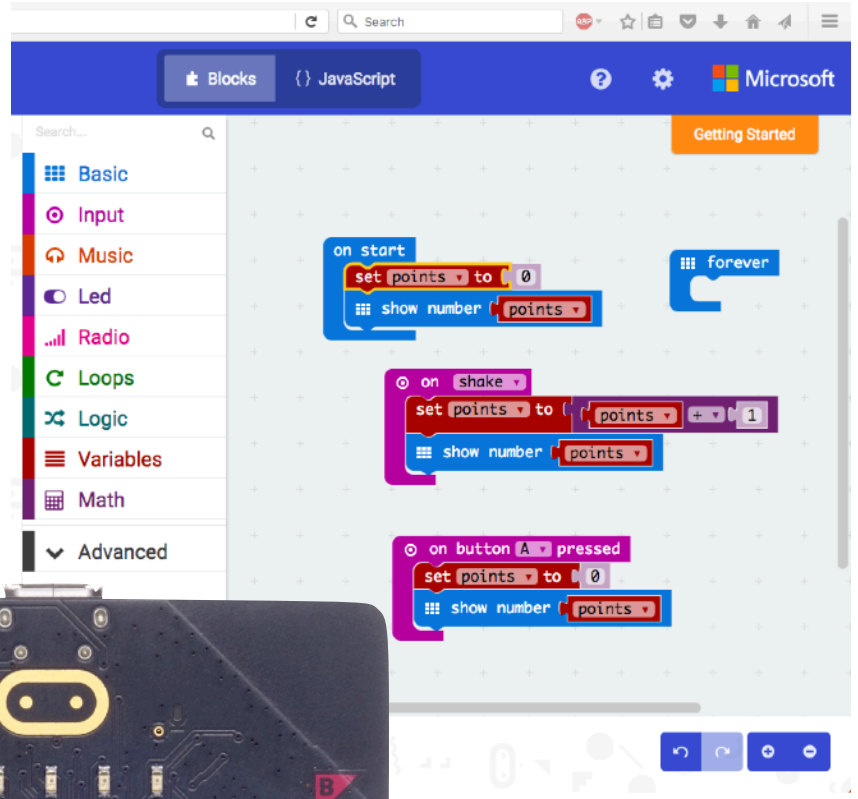
One practical project in one of:

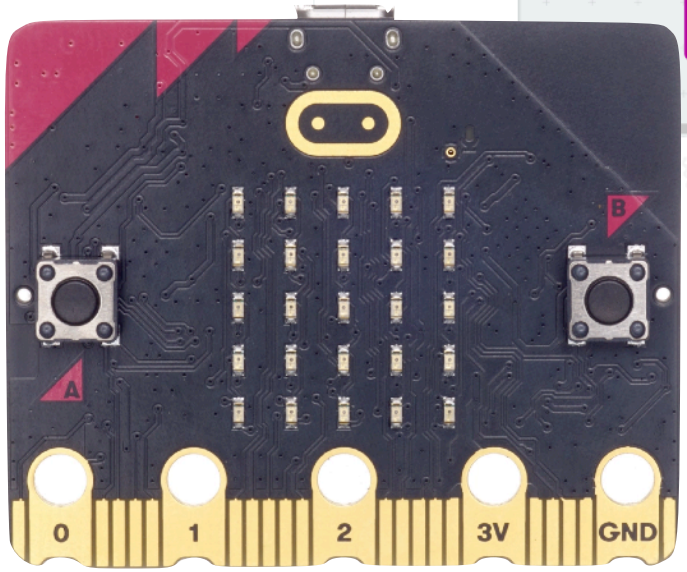
1. Biomechanics,
2. Imaging,
3. Health systems

In groups of 4-5 students. Project topic to be given by us!

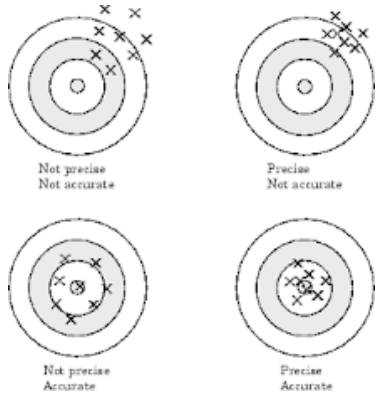
Lab 0 and Lab 1



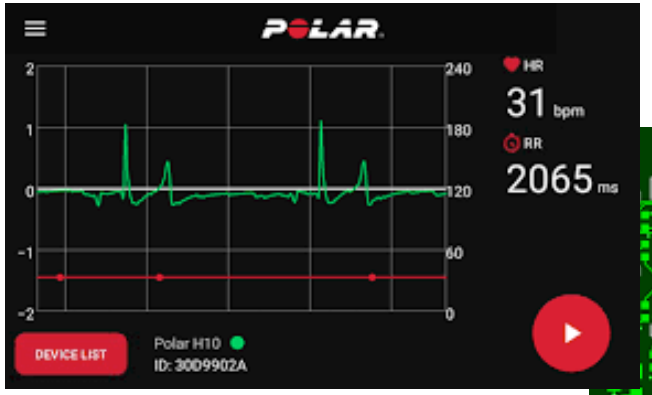


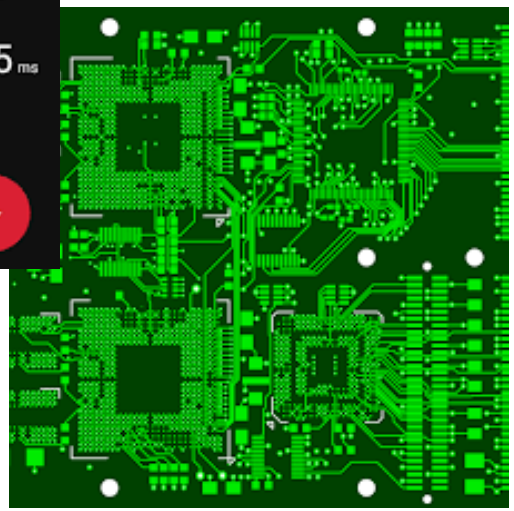


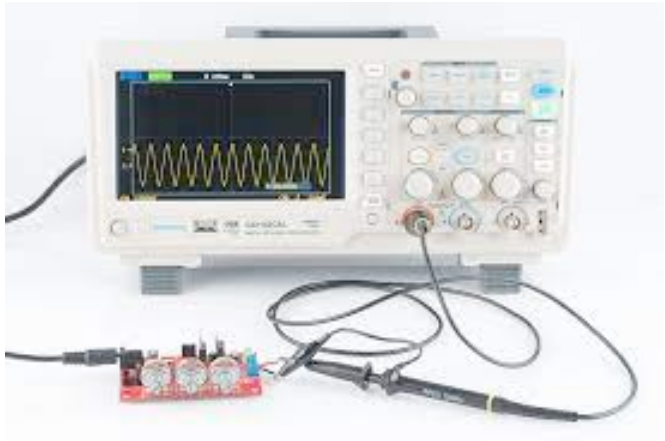
The remaining Labs



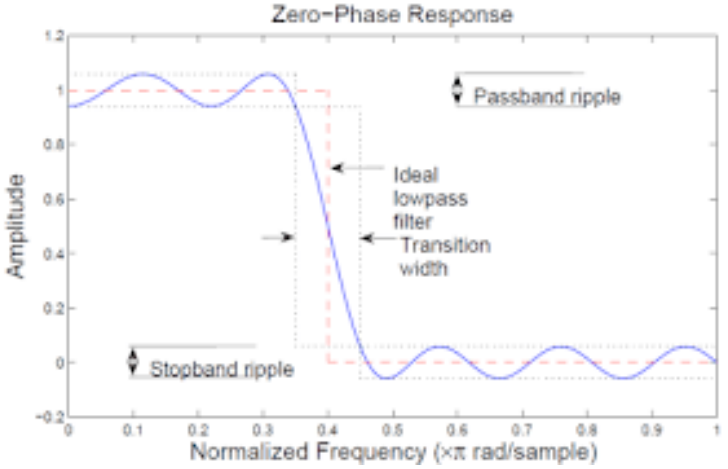
- Lab 2: PCB-design
- Lab 3: Soldering exercise
- Lab 4: Make a signal generator
- Lab 5: Digital filter design
- Lab 6: Implementation of a complete measurement system
- Lab 7: Basic measurements











[Maker]space is on floor 7

[Maker]space

- **You need access, we need your card number: Canvas Quiz**



https://canvas.kth.se/courses/19538

The Components



<u>The Kit:</u>

- Micro:Bit v2
- Breadboard
- Connector
- Cables
- ~~Bluetooth Dongle - *You don't need it!*~~
- Protoboard - for Lab 5

The rest found in MakerSpace!

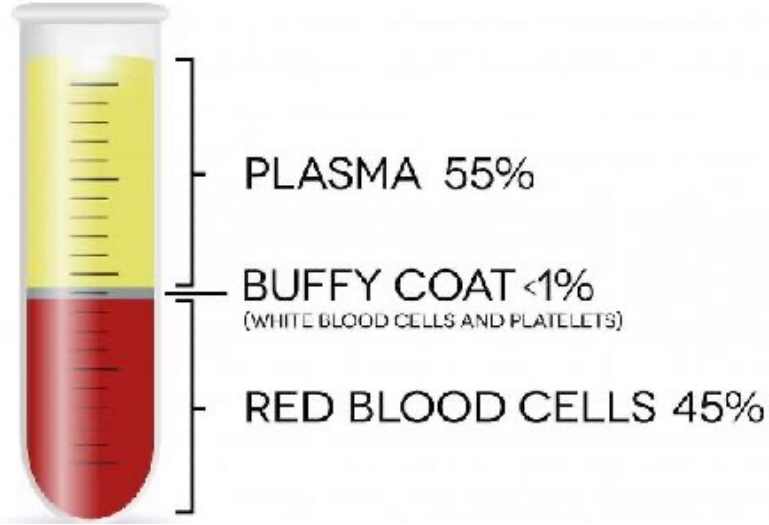


Medical Measurement Systems

Typical

Biomedical Measurements



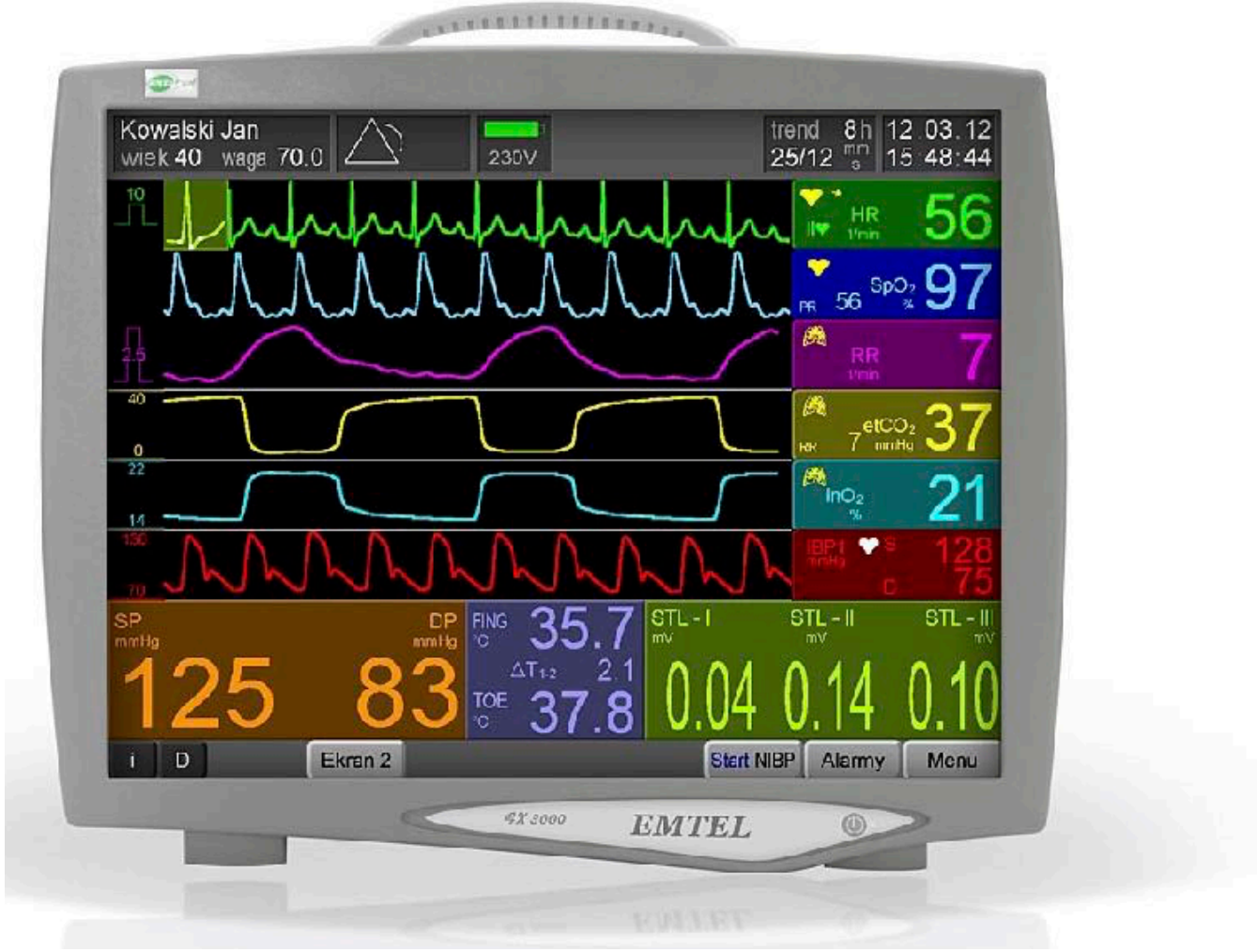




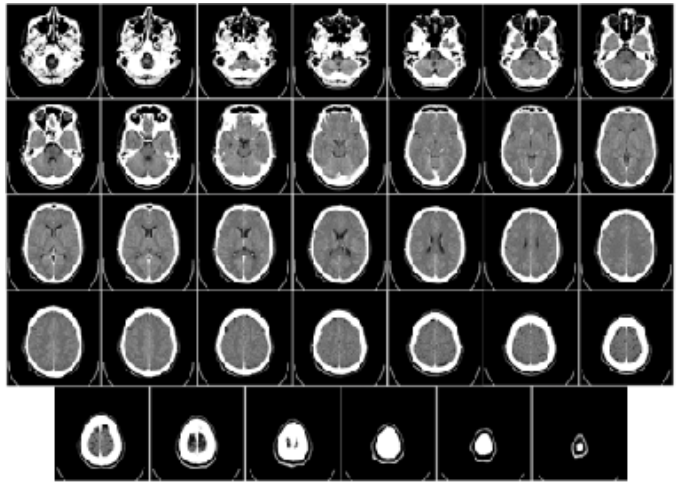
Biosignals for Patient Monitoring

Bedside Monitor
Vital signs, for use in

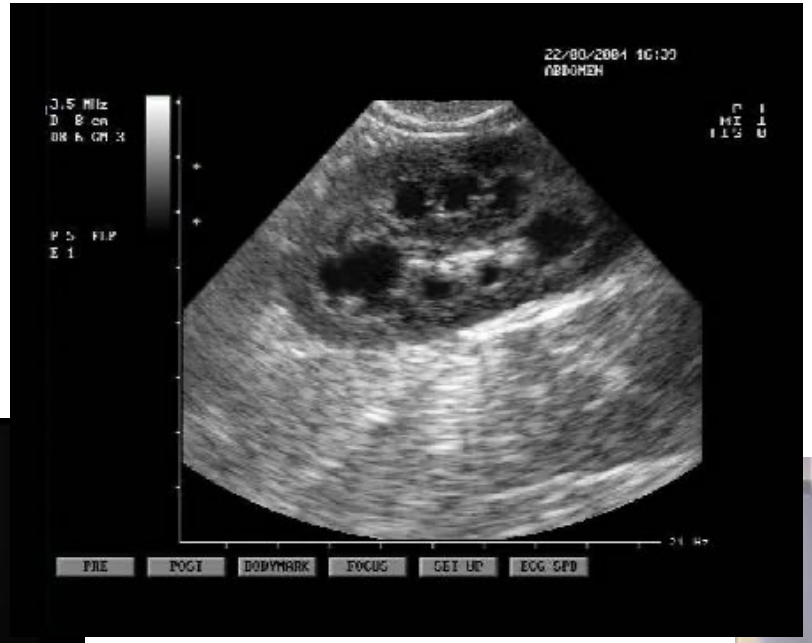
- an ICU
- An operation room
- pre/post operative ward
- etc.



Biomedical Imaging



CT Scan (X-Ray)



Ultrasound

X-ray





MRI



Medical Devices Applications

An application combines several tasks and functional blocks from different disciplines:

- Biomedical Instrumentation
- Signal Processing / Machine Learning
- Telematics
- Human Computer Interface

A Biomedical Instrumentation System

Transducer / sensor

Processing

Visualization



Measurand



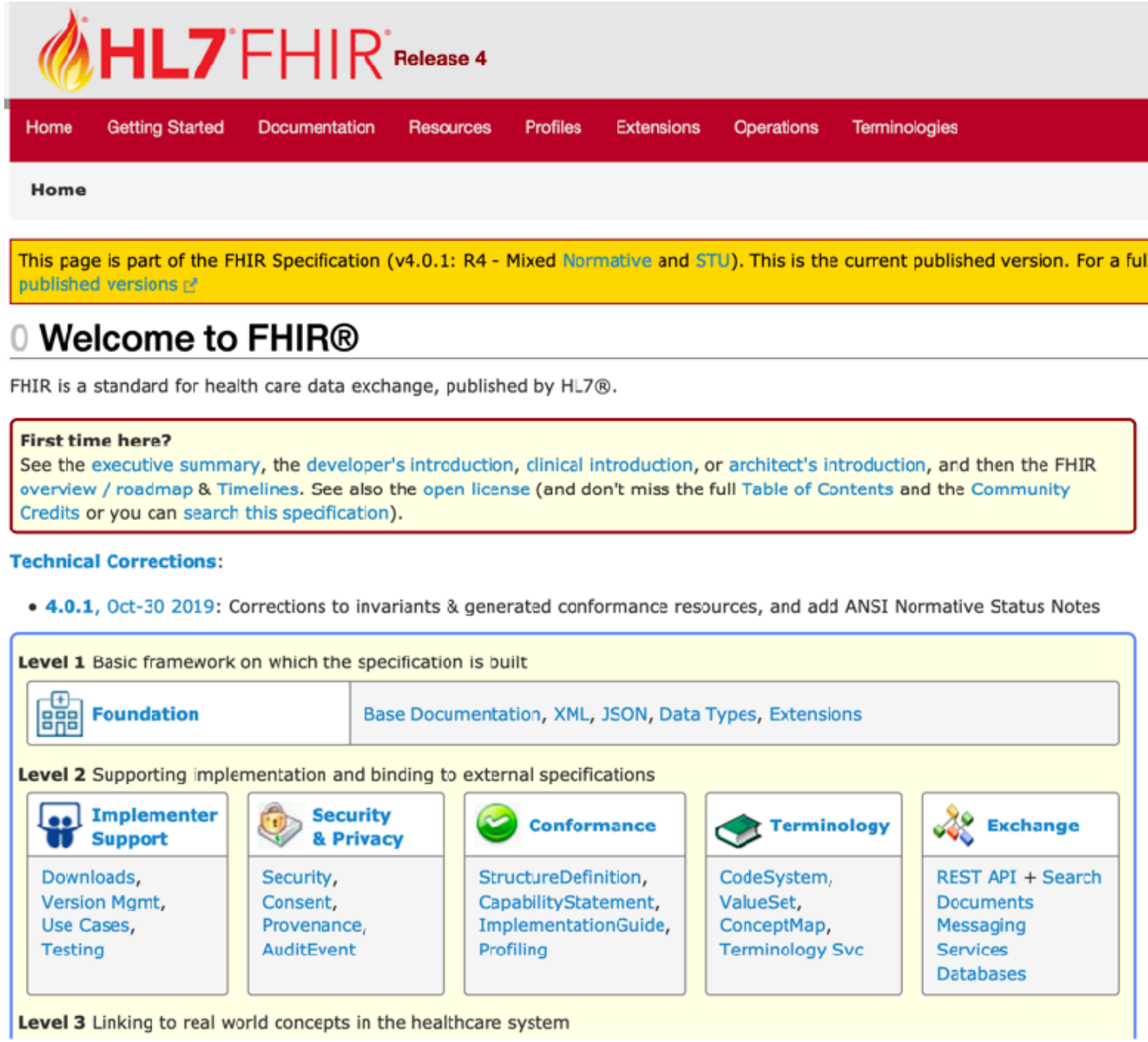


Element	Examples
Measurand	Blood pressure, EEG potential
Transducer	Cuff, EEG electrode
Processing	Feature extraction, filtering
Display	Pointer on a scale, polygraph

HL7 - FIHR

Fast Healthcare Interoperability Resources (FHIR)
is a standard describing data formats and elements (known as "resources") and an API for exchanging EHR.

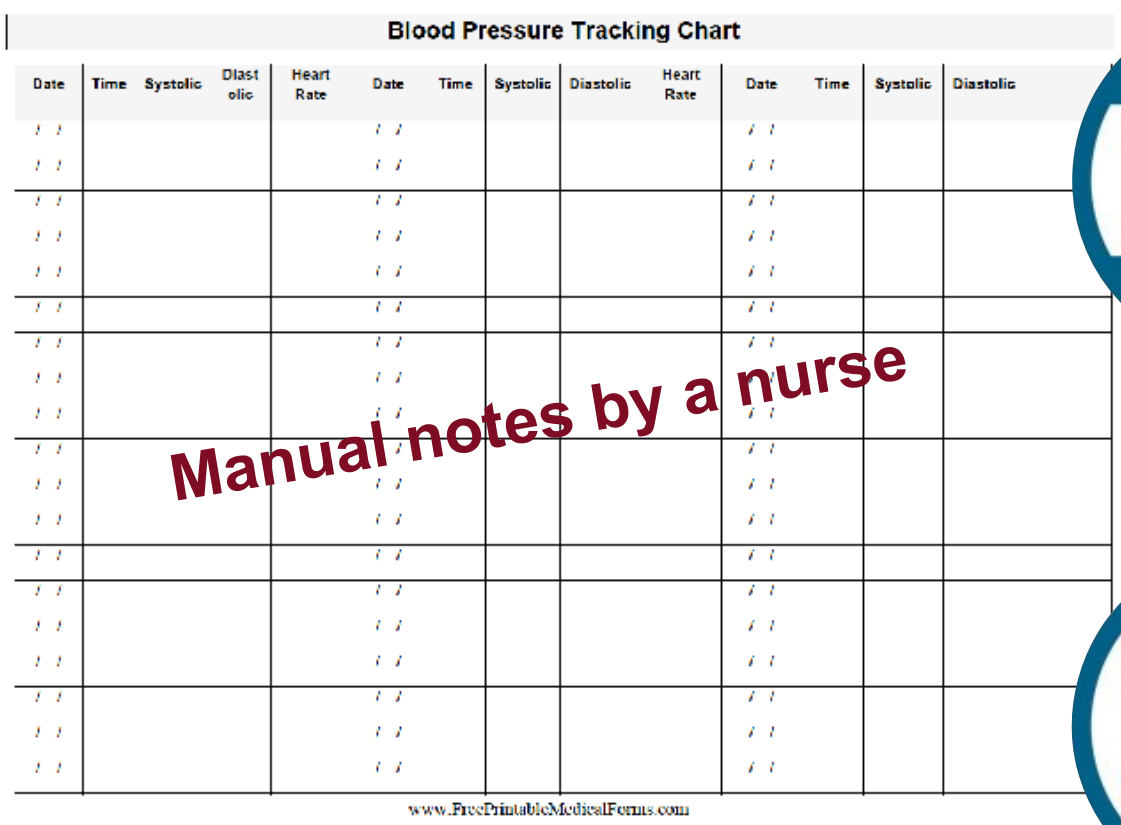
Based on HTTP(S), RESTful, HTML, CSS, JSON, XML, RDF, and/or Atom

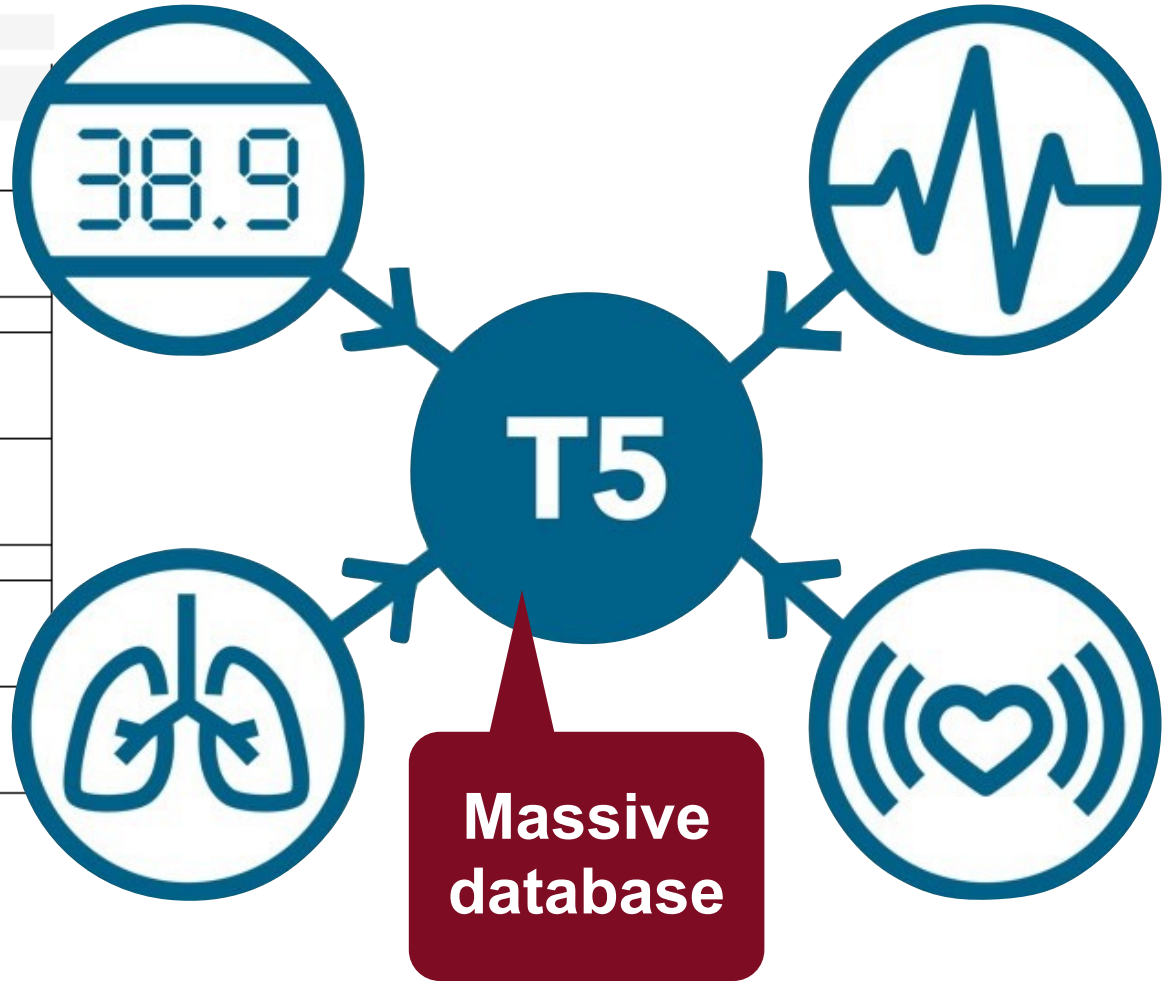


Name	Flags	Card.	Type	Description & Constraints
Observation	I N		DomainResource	Measurements and simple assertions + *Rule: dataAbsentReason SHALL only be present if Observation.value[x] is not present* + *Rule: If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present* Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Business Identifier for observation
basedOn	Σ	0..*	Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)	Fulfills plan, proposal or order
partOf	Σ	0..*	Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)	Part of referenced event
status	?! Σ	1..1	code	registered | preliminary | final | amended + ObservationStatus (Required)
category		0..*	CodeableConcept	Classification of type of observation Observation Category Codes (Preferred)
code	Σ	1..1	CodeableConcept	Type of observation (code / type) LOINC Codes (Example)
subject	Σ	0..1	Reference(Patient | Group | Device | Location)	Who and/or what the observation is about
focus	Σ TU	0..*	Reference(Any)	What the observation is about, when it is not about the subject of record
encounter	Σ	0..1	Reference(Encounter)	Healthcare event during which this observation is made
effective[x]	Σ	0..1		Clinically relevant time/time-period for observation
effectiveDateTime			dateTime	
effectivePeriod			Period	
effectiveTiming			Timing	
effectiveInstant			instant	
issued	Σ	0..1	instant	Date/Time this version was made available
performer	Σ	0..*	Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson)	Who is responsible for the observation

value[x]	Σ I	0..1		Actual result
valueQuantity			Quantity	
valueCodeableConcept			CodeableConcept	
valueString			string	
valueBoolean			boolean	
valueInteger			integer	
valueRange			Range	
valueRatio			Ratio	
valueSampledData			SampledData	
valueTime			time	
valueDateTime			dateTime	
valuePeriod			Period	
dataAbsentReason	I	0..1	CodeableConcept	Why the result is missing DataAbsentReason (Extensible)
interpretation		0..*	CodeableConcept	High, low, normal, etc. Observation Interpretation Codes (Extensible)
note		0..*	Annotation	Comments about the observation
bodySite		0..1	CodeableConcept	Observed body part SNOMED CT Body Structures (Example)
method		0..1	CodeableConcept	How it was done Observation Methods (Example)
specimen		0..1	Reference(Specimen)	Specimen used for this observation

Automatic Patient Data Collection





https://www.karolinska.se/om-oss/centrala-nyheter/2018/04/manga-vinnare-med-ny-hantering-av-patientens-viktigaste-matvarden/

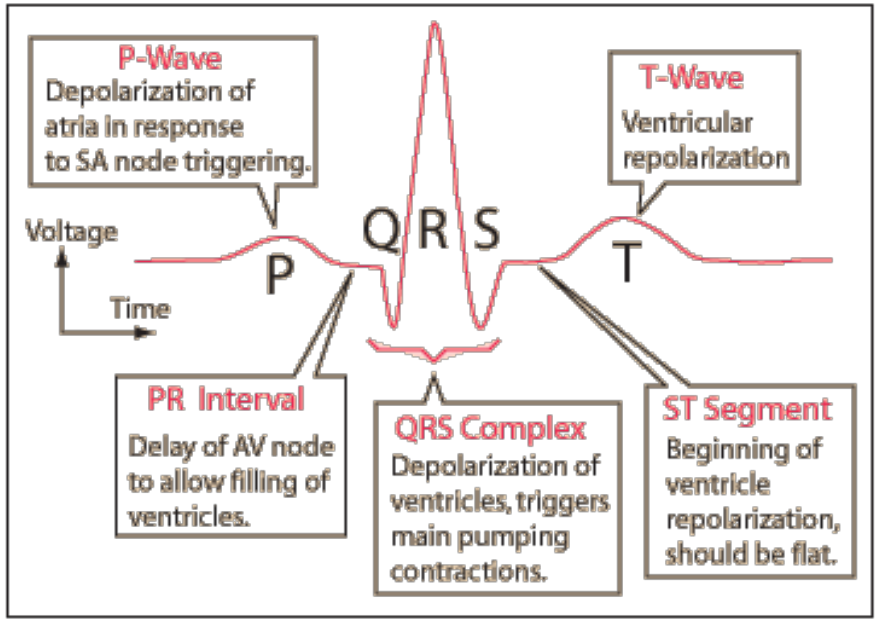
Typical Measurands (not complete!)

- Electrocardiogram (ECG)
- Electromyogram (EMG)
- Electroencephalogram (EEG)
- Phonocardiogram
- Electrogastrograph
- Nerve potentials
- Body Composition
- Posture
- Blood Oxygenation (SpO_2)
- Blood Pressure
- Blood Flow
- Heart Rate
- Bladder Pressure
- Temperature (Skin/Core)
- Respiration Rate
- Tidal Volume
- Stomach pH
- Intestinal forces

Typical Transducers / Sensors (not complete!)

- **Sensing Electrodes:** ECG, EMG, EEG, Electrogastrograph, Nerve potentials
- **Impedance Electrode:** Respiration rate, Body Composition
- **Glass Electrode:** Stomach pH
- **Microphone:** Phonocardiogram (Electronic Stethoscope)
- **Piezoelectric sensors:** Phonocardiogram, Respiration rate
- **Strain gauge:** Blood Pressure, Bladder Pressure, Intestinal Forces
- **Ultrasonic sensor:** Blood Flow, Body Composition (Ultrasound Imaging)
- **Impedance Pneumograph:** Tidal Volumen
- **Optic Fibre:** Respiration Rate
- **Infrared sensors:** Pulse, Blood Oxygenation ($Sp0_2$)
- **Coils antenna:** Body composition (MR)
- **Film:** Body composition (X-Ray)

Electrocardiogram (ECG/EKG)





Endogenic cardiac electrical activity

Multiple skin-mounted electrodes:

0.1 - 5 mV 0.5 - 40 Hz

Example medical usage:

- Find heart diseases
- Inflammation around the heart
- Check pacemakers
- Monitor drug effects
- Cardiac stress test
- Vital signs monitoring

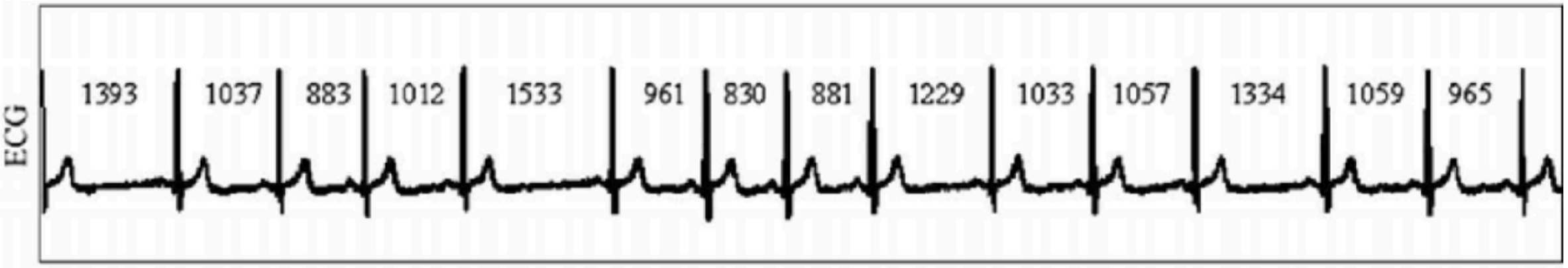
Heart Rate and HRV

- Time interval between two QRS pulses
- Heart rate: 60 seconds / RR time interval
- Beats per minute BPM: 50-180
- Heart Rate Variability (HRV)
- Obtained from ECG, blood pressure, and/or SpO_2

Should be “lagom” HRV.

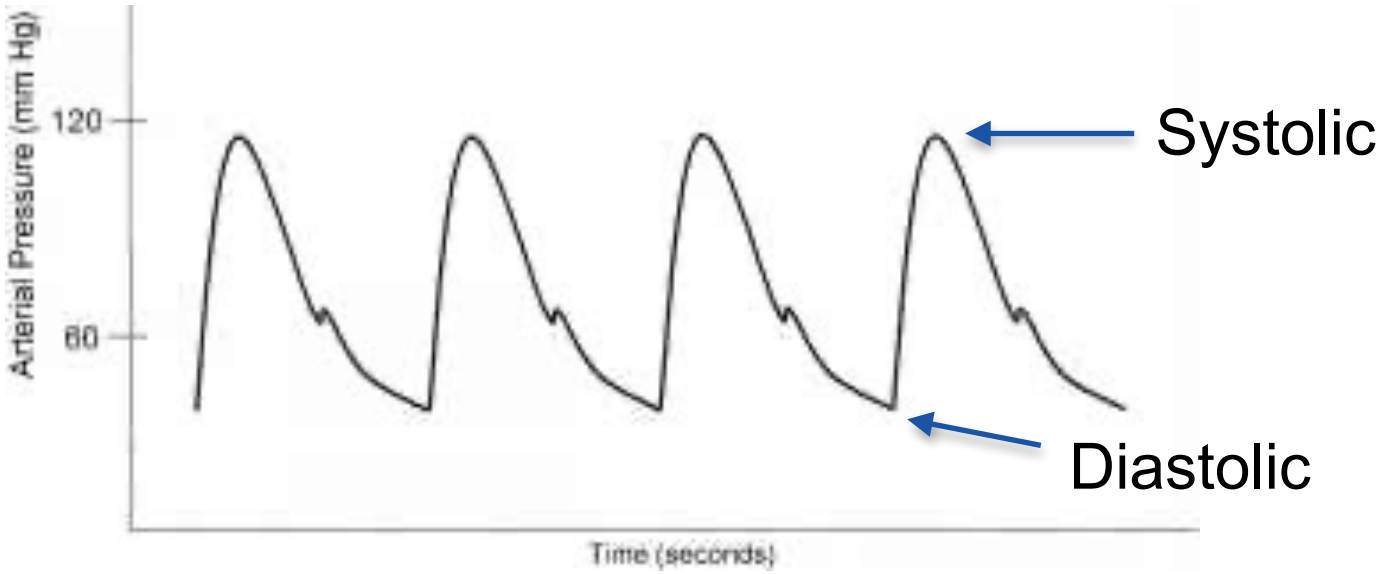
Arrhythmia!
e.g. atrial fibrillation
Förmaksflimmer

RR Interval times in milliseconds



Blood Pressure

- Continuous pressure measurements
- Several places: Arterial Systemic, Venous, Pulmonary, Ventricular, …
- Several values derived from the blood pressure waveform (SBP, DBP, MAP, …)



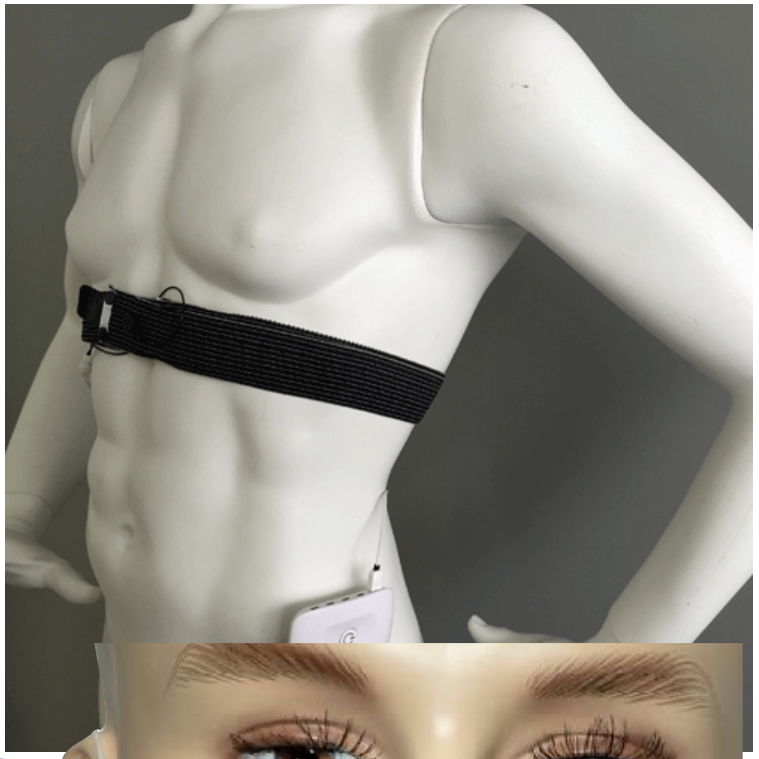




Respiration Rate

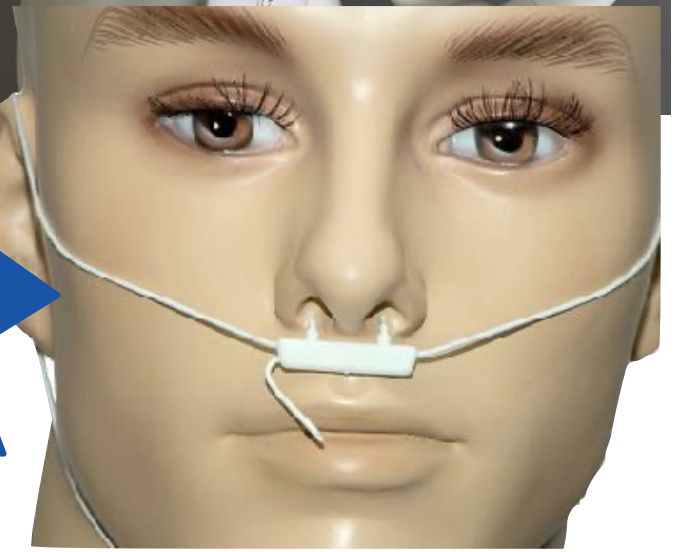
- Spirometry
- Resistive gauge
- Nose / mouth thermistor (temperature)
- Nose / mouth optic sensor (humidity)
- Impedance pneumography
- WiFi radio (MIT Vital-Radio)
- Microphone (sound)

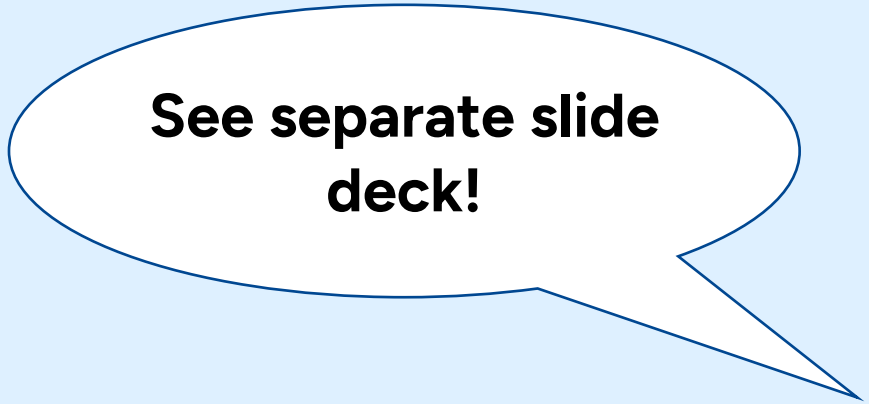






Multiple Sensors Possible!



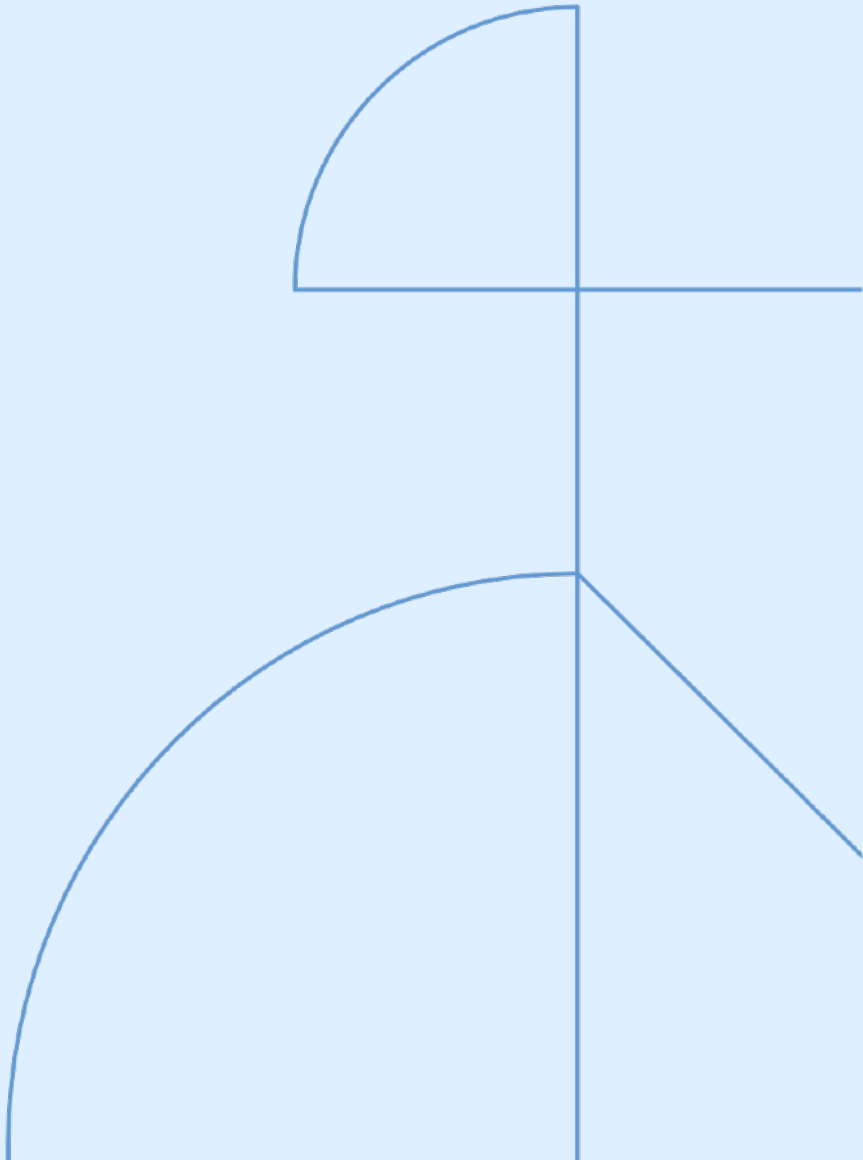


Lab 0

See Lab 0 slides

Lab 1

Program a Medical Device



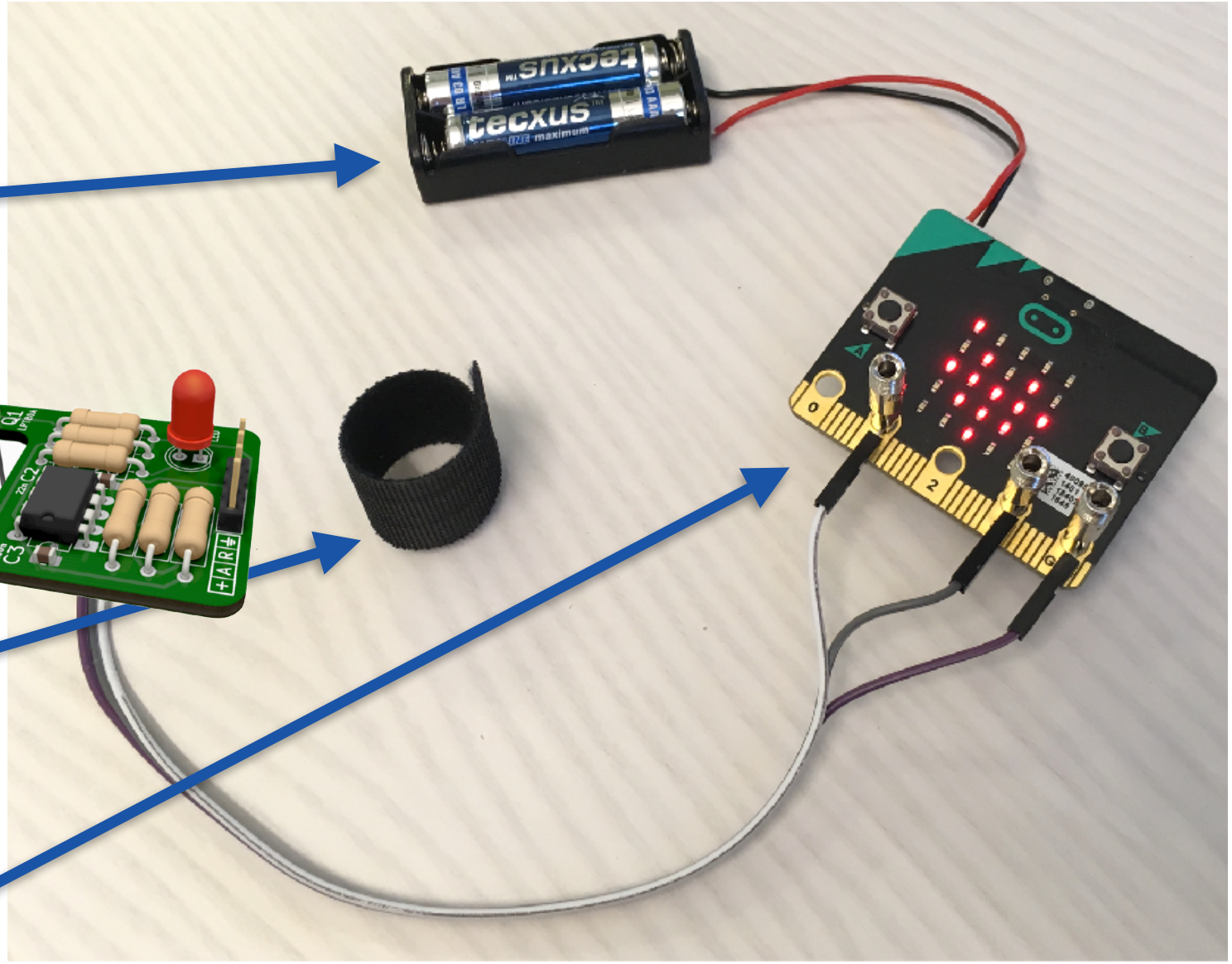
Hardware

Battery Pack
(Not needed)

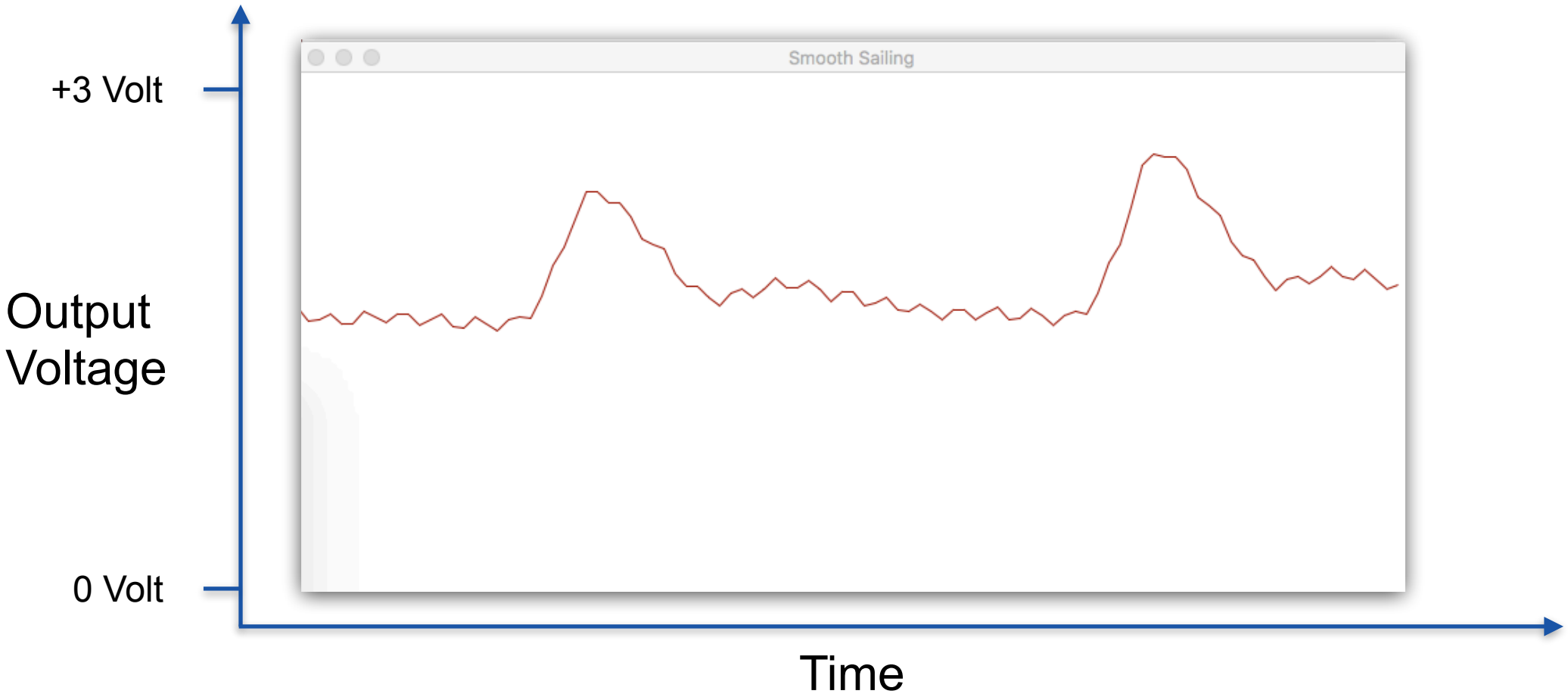
photoplethysmograph

Velcro tape
(To stick the sensor to the finger)

Processing Unit



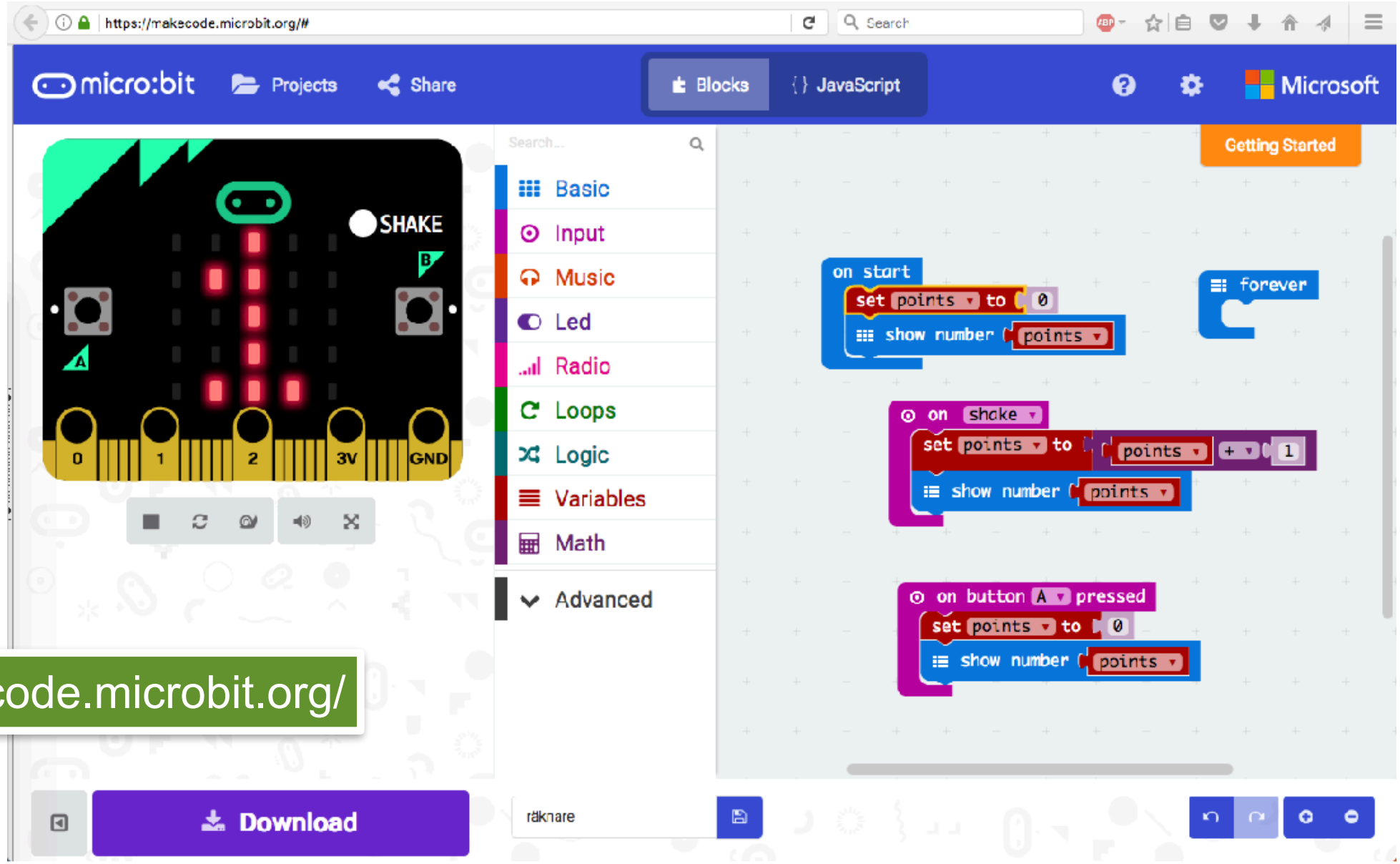
Graph Drawing



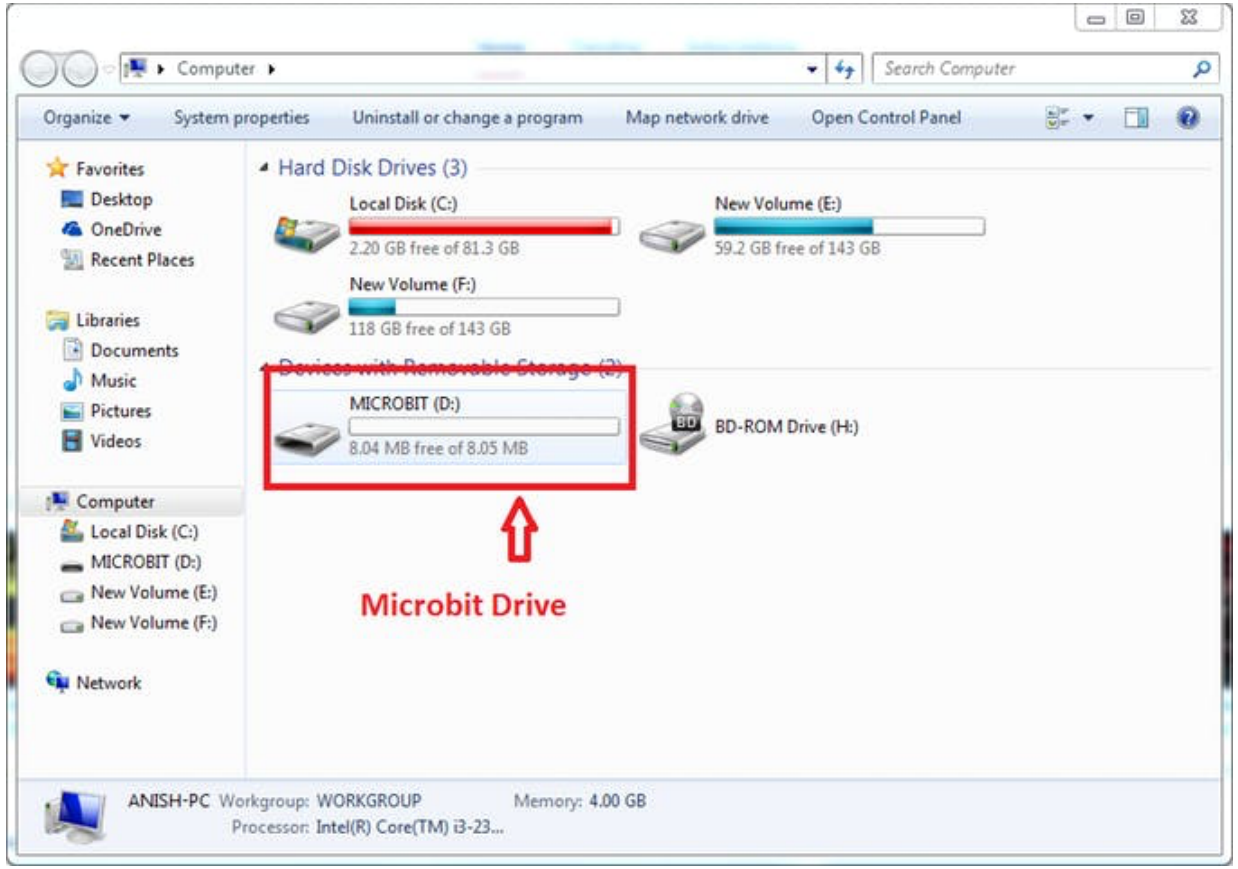
Step 1:
Test the Development Environment

Try the simulator

http://makecode.microbit.org/



Step 2: Program the Micro:bit



Check that the Micro:bit drive appears

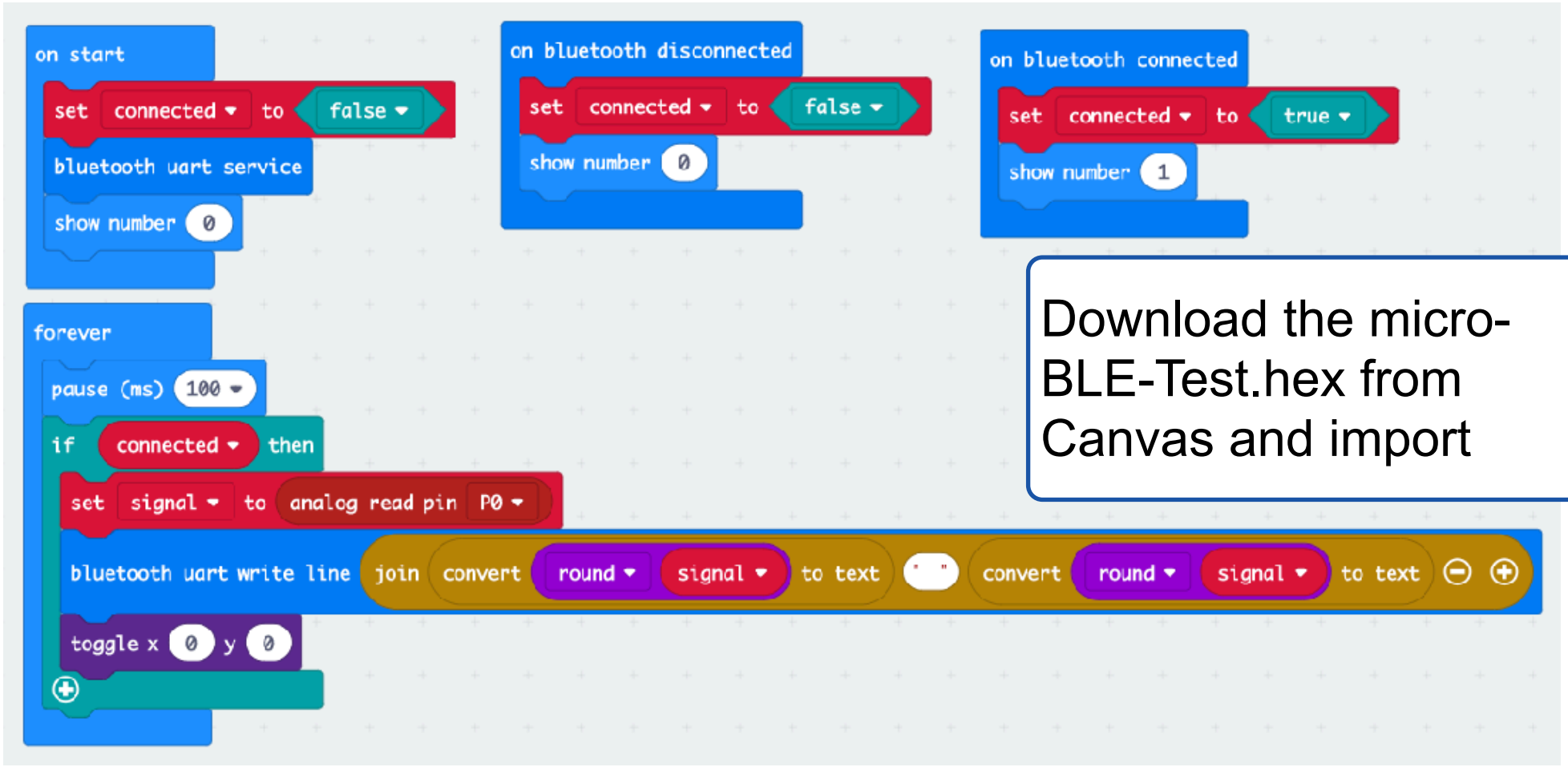
Step 2: Program the Micro:bit (cont:d)

1. Connect it to the computer with the USB cable.
2. Wait.
3. Maybe you are asked to install updates. Do that.
4. Change the setting in Chrome:
    1. Click  in the top right corner
    2. Choose Settings / Advanced / Download
    3. Choose Ask where to save.
5. Go back to Development Environment
6. Click Download and save to the Micro:bit drive.

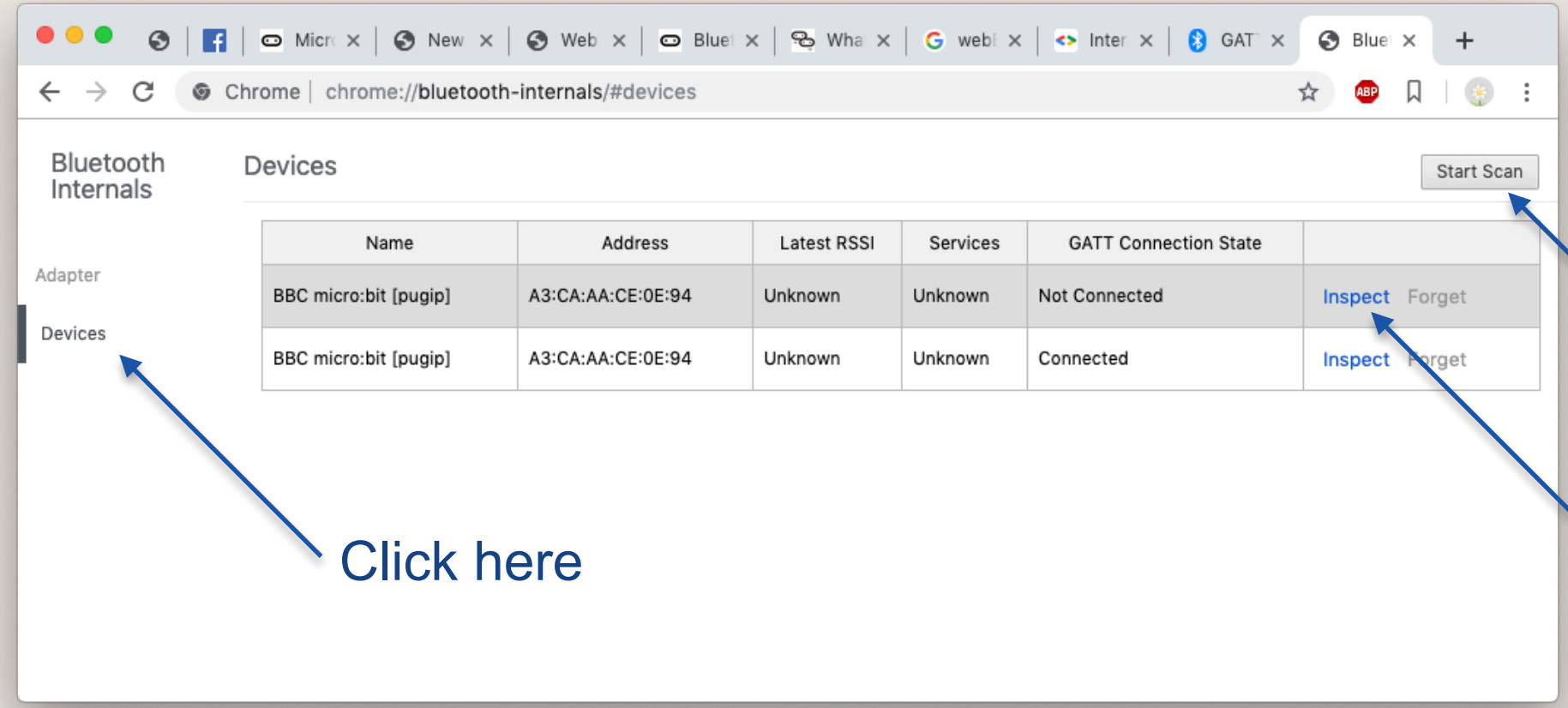


7. Test the code on the Micro:bit by shaking it and press button A.

A basic BLE program for the Micro:bit



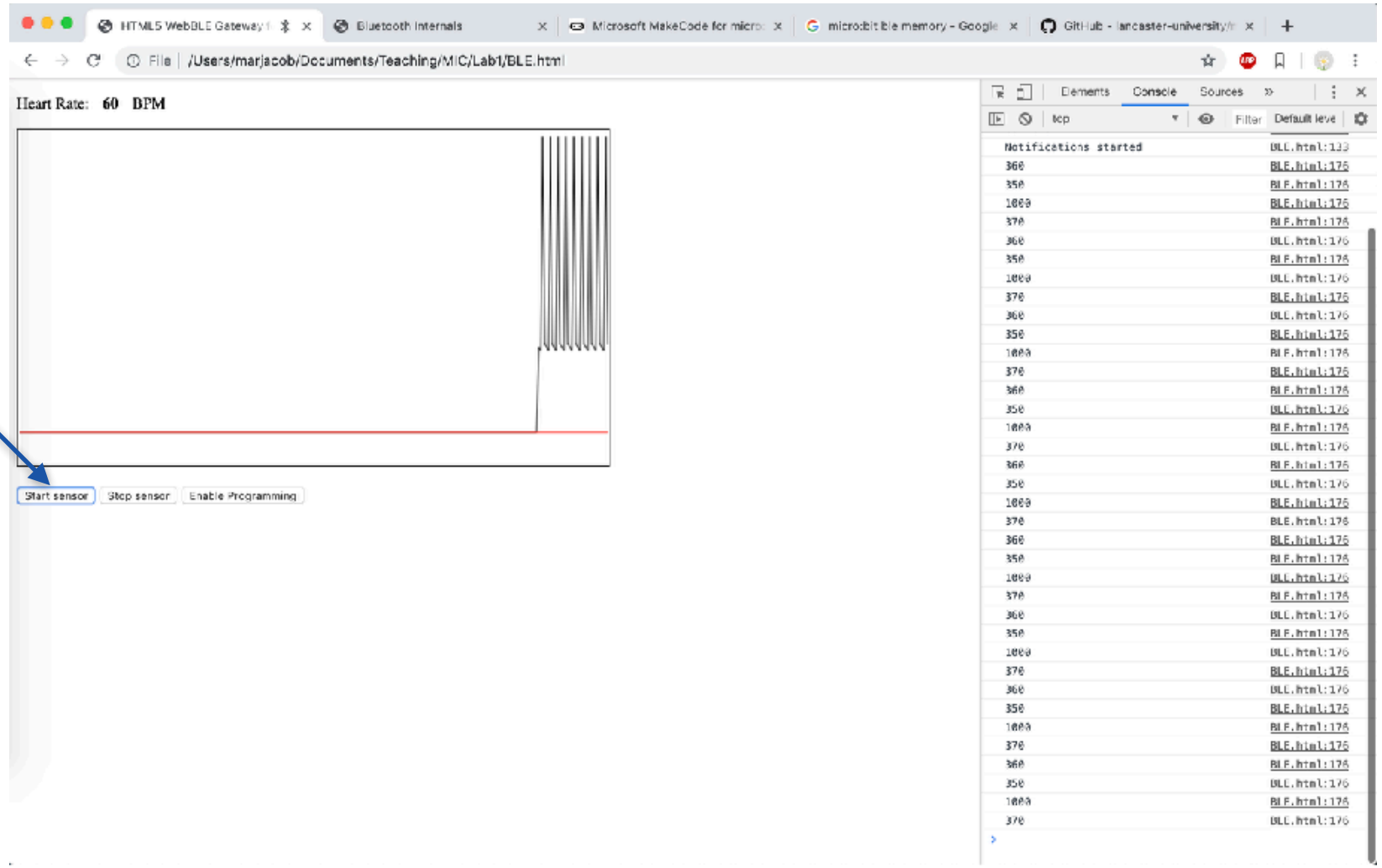
WebBLE


Chrome
chrome://bluetooth-internals/#devices
Bluetooth Internals
Devices
Start Scan
Adapter
Devices
Name
Address
Latest RSSI
Services
GATT Connection State
BBC micro:bit [pugip]
A3:CA:AA:CE:0E:94
Unknown
Unknown
Not Connected
Inspect
Forget
BBC micro:bit [pugip]
A3:CA:AA:CE:0E:94
Unknown
Unknown
Connected
Inspect
Forget
Click here
Click here
Click here

Connect to the Web page in Chrome

Click here to connect to the device

Don’t open file in Canvas! Download webpage and open in Chrome



Step 6: Make the Code

- Use the pulsesensor and the Micro:bit

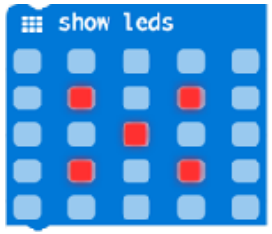
- Make a device that can measure the heart beat (BPM)
- Blink a LED with the heart beat
- Connect it to a computer with Bluetooth, and
- Calculate the BPM and display it with the WebBLE web script in Google Chrome
- together with the blood vessel graph.

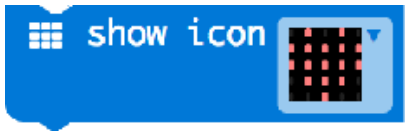
Some Small Hints

Find the ‘time’:



Some commands are slow, such as:





Instead use:



(found under Led)

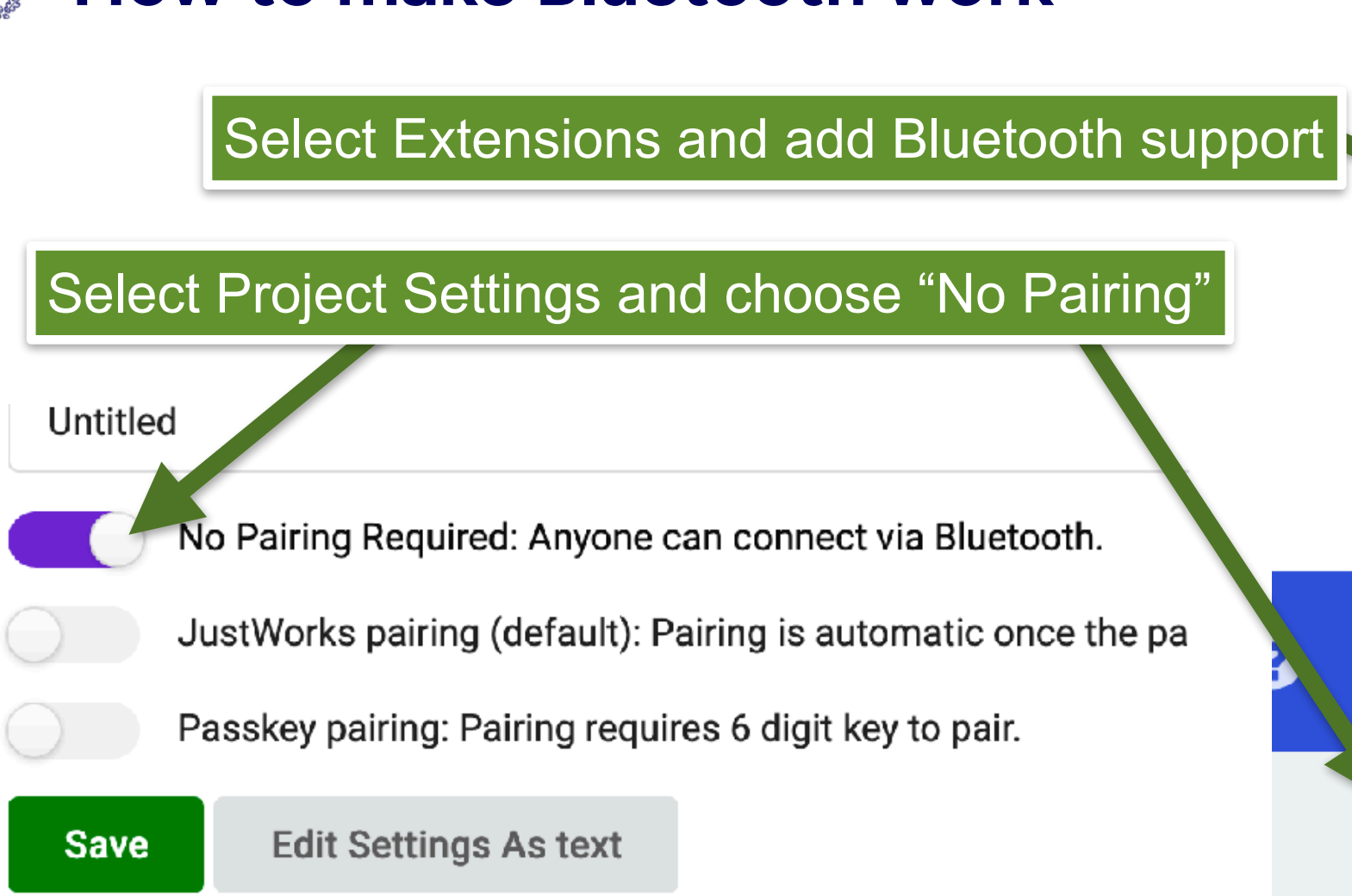
Float math is problematic:

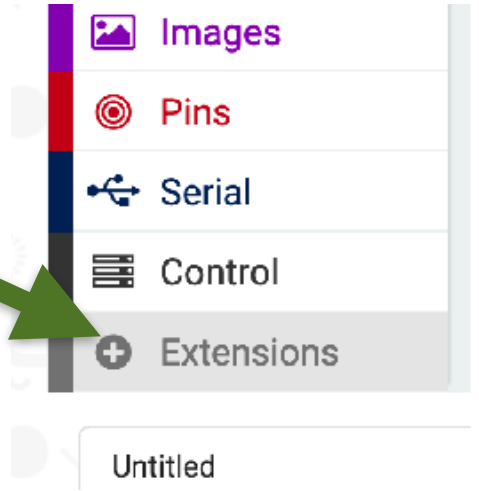


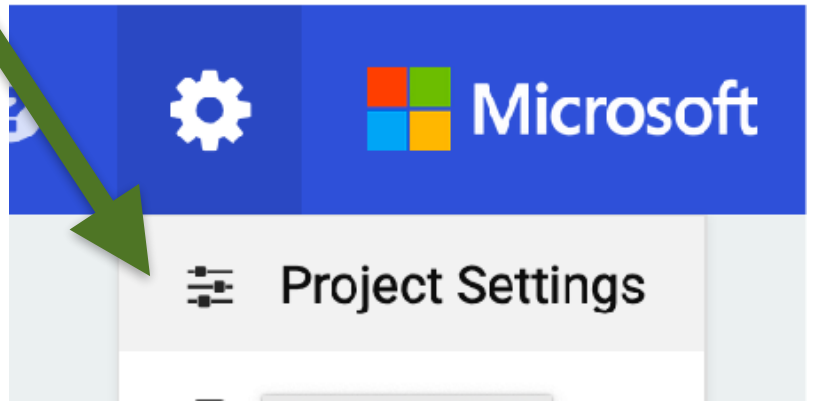
How to make Bluetooth work

Select Extensions and add Bluetooth support

Select Project Settings and choose “No Pairing”

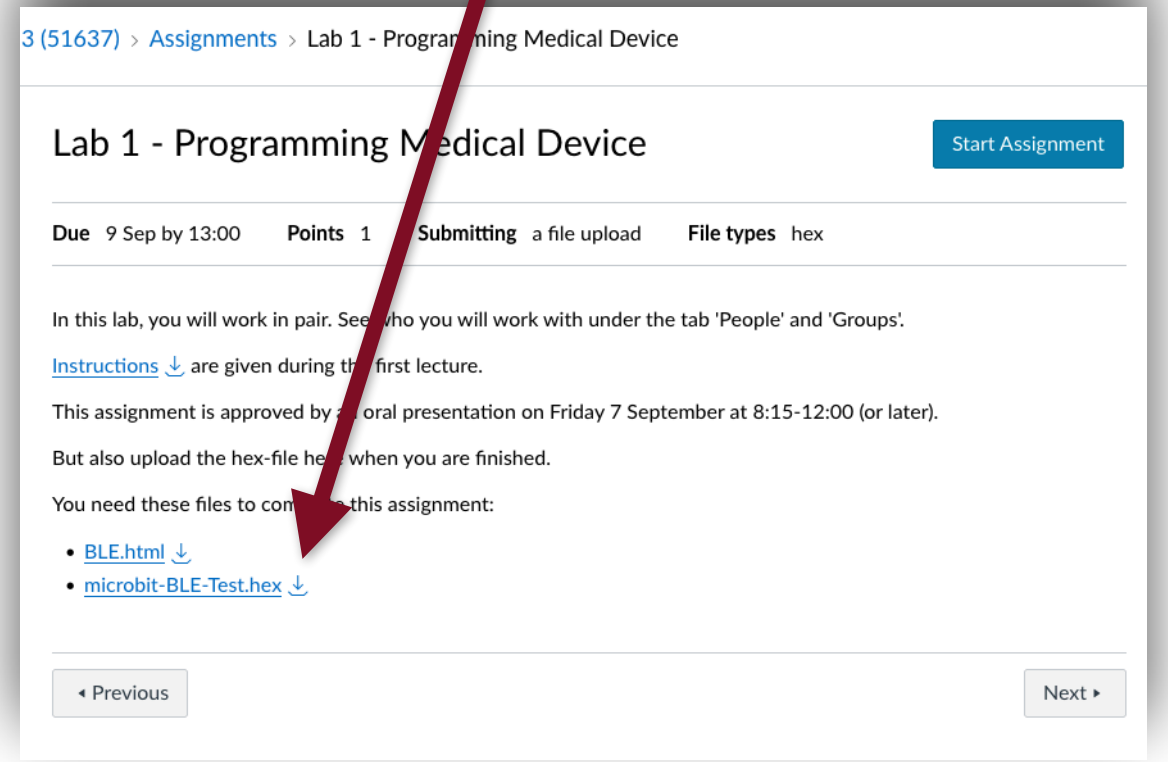


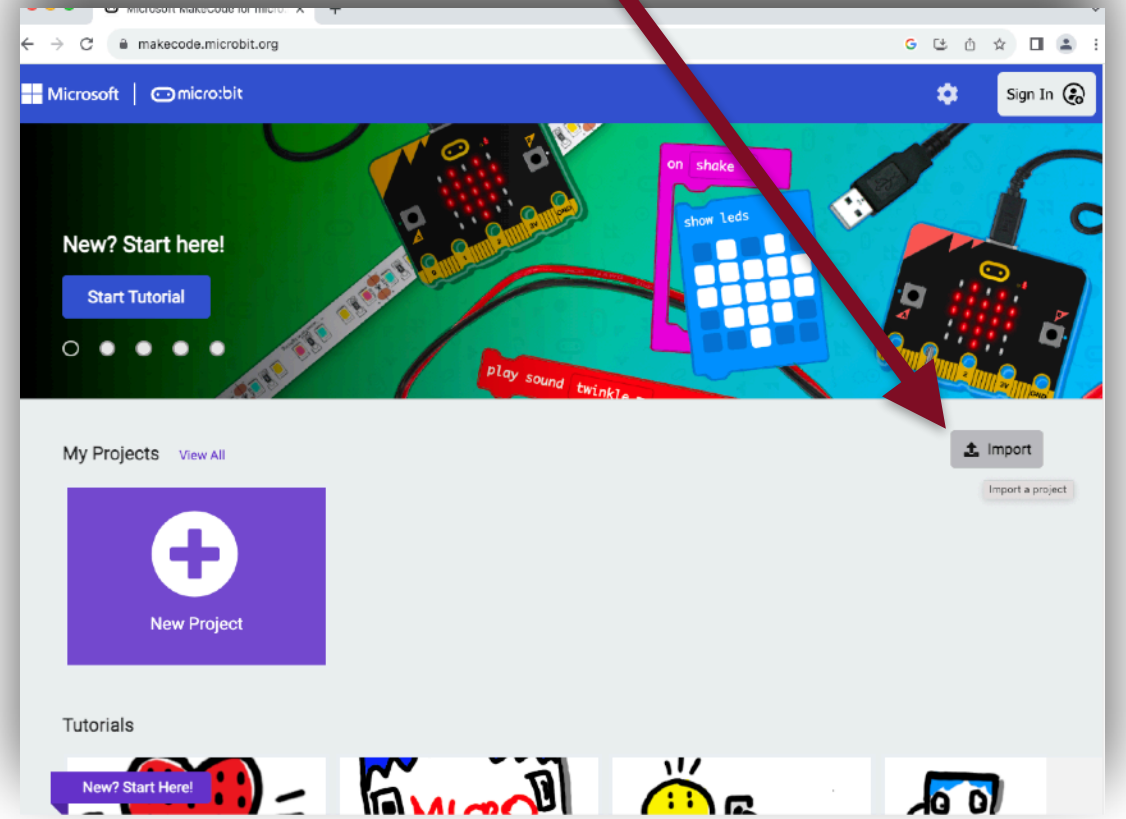




If previous slide does not work...

Download the hex-file from KTH Canvas and import!






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VETENSKAP
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